No.	Status	Date
HIS-2011-0207	Issued	2011-02-18

UT Thickness Survey - ATU-100 Sour Gas and Rich Amine

Scope and Responsibilities

This report details the 5-year Class 1 survey for the ATU-100 at the Hamilton Plant. The scope includes ultrasonic thickness measurements of sour gas inlet lines 6-SG-101 and 6-SG-102, and rich amine lines 4-RA-101, 4-RA-102, 6-RA-103, and 4-RA-104. The objective is to verify mechanical integrity and calculate corrosion rates in accordance with API 570 and ASME B31.3 requirements.

Kevin Doucette served as the lead UT technician, performing all field measurements and data recording. Anil Rao provided technical oversight and final review of the inspection data. All activities were conducted under the quality management framework of Halward Inspection Services Ltd. to ensure compliance with site-specific asset integrity standards.

Inspection Procedures and Standards

Field work was performed under QSCP-NDT-001 Rev 2, which incorporates recent revisions for the CUI program including profile radiography and insulation management. Measurements were executed following HIS-UT-002 Rev 1, which has been accepted by the QSCP Hamilton Inspection Department. The methodology for straight-beam ultrasonic examination of contact thickness follows ASTM E797.

Personnel qualifications for Kevin Doucette and Anil Rao are maintained in accordance with ASNT SNT-TC-1A. The inspection strategy aligns with QSCP-IP-ATU100 Rev 2, which mandates that insulated lines operating within the CUI range receive profile RT or insulation windows every 5 years to mitigate external corrosion risks.

Historical Data Comparison

Current thickness data was compared against HIS-2006-0412, which documented the previous 10-year full-unit survey. For the 4-RA-104 spool, the baseline data established in HIS-2010-0318 was utilized to determine the short-term corrosion rate. These historical records provide the basis for calculating the long-term metal loss trends across the amine treating circuit.

The 3-year regenerator overhead survey cycle, as noted in HIS-2010-0318, remains independent of this 5-year Class 1 cycle. However, the data from both reports are integrated into the plant's master records to ensure all Class 1, 2, and 3 piping systems meet the required thickness intervals.

Analysis of Sour Gas Lines

Lines 6-SG-101 and 6-SG-102, both A106 Gr.B with a nominal thickness of 7.11 mm, were inspected at all designated CMLs. With a design pressure of 1965 kPa(g) and a design temperature of 66 °C, the retirement thickness is set at a t-min of 3.56 mm. Current readings indicate that these sour gas inlets remain well above the retirement threshold.

Corrosion rates for these sections were calculated using the larger of the long-term and short-term rates. Based on the current metal loss trends, the remaining life for 6-SG-101 and 6-SG-102 exceeds the 5-year maximum interval for Class 1 service. No NCRs were issued for these circuits during this survey.

Analysis of Rich Amine Lines

The rich amine header 6-RA-103 and lines 4-RA-101 and 4-RA-102 were evaluated against their respective t-min values. For 4-RA-101 and 4-RA-102, the t-min is 3.81 mm against a nominal 8.56 mm. Line 6-RA-103, with a nominal 10.97 mm, has a t-min of 4.57 mm. These lines operate under design conditions of 862 kPa(g) and 82 °C.

Line 4-RA-104, which services the rich side of the L/R exchanger at a design temperature of 127 °C, was also surveyed. Data analysis shows that the rich amine circuit has reached half of its remaining life. Per site rules, the next thickness reading must be scheduled at no more than half the remaining life, which will occur before the standard 10-year Class 2 maximum.

Thickness readings, measured 2011-02-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2006-02-24	6.59	6.39	0.049	0.040	>50	2016-02-18	
6-SG-101-02	7.11	3.56	2006-02-24	6.51	6.14	0.061	0.074	34.7	2016-02-18	
6-SG-101-03	7.11	3.56	2006-02-24	6.42	6.17	0.061	0.050	42.9	2016-02-18	
6-SG-101-04	7.11	3.56	2006-02-24	6.28	6.05	0.070	0.046	35.8	2016-02-18	
6-SG-101-05	7.11	3.56	2006-02-24	6.22	5.87	0.085	0.070	27.2	2016-02-18	
6-SG-101-06	7.11	3.56	2006-02-24	6.49	6.27	0.054	0.044	>50	2016-02-18	
6-SG-101-08	7.11	3.56	—	—	6.33	—	—	—	2016-02-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2006-02-24	6.26	5.75	0.093	0.102	21.4	2016-02-18	
6-SG-102-02	7.11	3.56	2006-02-24	6.25	5.80	0.088	0.090	24.8	2016-02-18	
6-SG-102-03	7.11	3.56	2006-02-24	6.26	5.80	0.083	0.092	24.3	2016-02-18	
6-SG-102-04	7.11	3.56	2006-02-24	6.22	5.73	0.084	0.098	22.1	2016-02-18	
6-SG-102-05	7.11	3.56	2006-02-24	6.72	6.46	0.045	0.052	>50	2016-02-18	
6-SG-102-06	7.11	3.56	—	—	6.41	—	—	—	2016-02-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2006-02-24	8.27	8.32	0.013	-0.010	>50	2021-02-18	
4-RA-101-02	8.56	3.81	2006-02-24	8.27	8.20	0.023	0.014	>50	2021-02-18	
4-RA-101-03	8.56	3.81	2006-02-24	8.10	7.98	0.042	0.024	>50	2021-02-18	
4-RA-101-04	8.56	3.81	2006-02-24	8.34	8.21	0.028	0.026	>50	2021-02-18	
4-RA-101-05	8.56	3.81	2006-02-24	8.29	8.16	0.029	0.026	>50	2021-02-18	
4-RA-101-07	8.56	3.81	2006-02-24	8.30	8.30	0.020	0.000	>50	2021-02-18	
4-RA-101-08	8.56	3.81	2006-02-24	8.07	8.02	0.044	0.010	>50	2021-02-18	
4-RA-101-09	8.56	3.81	2006-02-24	8.43	8.32	0.017	0.022	>50	2021-02-18	
4-RA-101-10	8.56	3.81	2006-02-24	8.27	8.28	0.026	-0.002	>50	2021-02-18	
4-RA-101-11	8.56	3.81	2006-02-24	8.14	8.04	0.043	0.020	>50	2021-02-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2006-02-24	8.41	8.33	0.014	0.016	>50	2021-02-18	
4-RA-102-02	8.56	3.81	2006-02-24	8.15	7.99	0.039	0.032	>50	2021-02-18	
4-RA-102-03	8.56	3.81	2006-02-24	8.27	8.04	0.035	0.046	>50	2021-02-18	
4-RA-102-04	8.56	3.81	2006-02-24	8.24	7.96	0.039	0.056	>50	2021-02-18	
4-RA-102-05	8.56	3.81	2006-02-24	8.19	8.07	0.032	0.024	>50	2021-02-18	
4-RA-102-06	8.56	3.81	2006-02-24	8.15	8.01	0.033	0.028	>50	2021-02-18	
4-RA-102-07	8.56	3.81	2006-02-24	8.29	8.10	0.032	0.038	>50	2021-02-18	
4-RA-102-08	8.56	3.81	2006-02-24	8.05	7.88	0.041	0.034	>50	2021-02-18	
4-RA-102-09	8.56	3.81	2006-02-24	8.18	7.98	0.038	0.040	>50	2021-02-18	
4-RA-102-11	8.56	3.81	—	—	8.19	—	—	—	2021-02-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2006-02-24	10.76	10.47	0.035	0.058	>50	2021-02-18	
6-RA-103-02	10.97	4.57	2006-02-24	10.50	10.23	0.058	0.054	>50	2021-02-18	
6-RA-103-03	10.97	4.57	2006-02-24	10.67	10.42	0.038	0.050	>50	2021-02-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-04	10.97	4.57	2006-02-24	10.40	9.97	0.068	0.086	>50	2021-02-18	
6-RA-103-05	10.97	4.57	2006-02-24	10.34	10.10	0.057	0.048	>50	2021-02-18	
6-RA-103-06	10.97	4.57	2006-02-24	10.74	10.56	0.030	0.036	>50	2021-02-18	
6-RA-103-07	10.97	4.57	2006-02-24	10.66	10.55	0.029	0.022	>50	2021-02-18	
6-RA-103-08	10.97	4.57	2006-02-24	10.58	10.44	0.035	0.028	>50	2021-02-18	
6-RA-103-09	10.97	4.57	2006-02-24	10.67	10.56	0.030	0.022	>50	2021-02-18	
6-RA-103-10	10.97	4.57	2006-02-24	10.27	10.05	0.065	0.044	>50	2021-02-18	
6-RA-103-13	10.97	4.57	—	—	10.45	—	—	—	2021-02-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2006-02-24	8.16	7.95	0.037	0.042	>50	2021-02-18	
4-RA-104-02	8.56	3.81	2006-02-24	8.23	8.14	0.027	0.018	>50	2021-02-18	
4-RA-104-03	8.56	3.81	2010-09-13	8.59	8.56	0.069	0.069	>50	2021-02-18	
4-RA-104-04	8.56	3.81	2010-09-13	8.52	8.60	-0.185	-0.185	>50	2021-02-18	
4-RA-104-05	8.56	3.81	2010-09-13	8.43	8.57	-0.324	-0.324	>50	2021-02-18	
4-RA-104-06	8.56	3.81	2006-02-24	8.09	7.98	0.041	0.022	>50	2021-02-18	
4-RA-104-07	8.56	3.81	2006-02-24	7.97	7.78	0.054	0.038	>50	2021-02-18	
4-RA-104-08	8.56	3.81	—	—	8.18	—	—	—	2021-02-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2011-02-18	_____
Reviewed	Anil Rao	Level III, SNT-TC-1A UT/RT Level III	2011-02-18	_____